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EPSRC

An entangled-LED driven quantum relay over 1km

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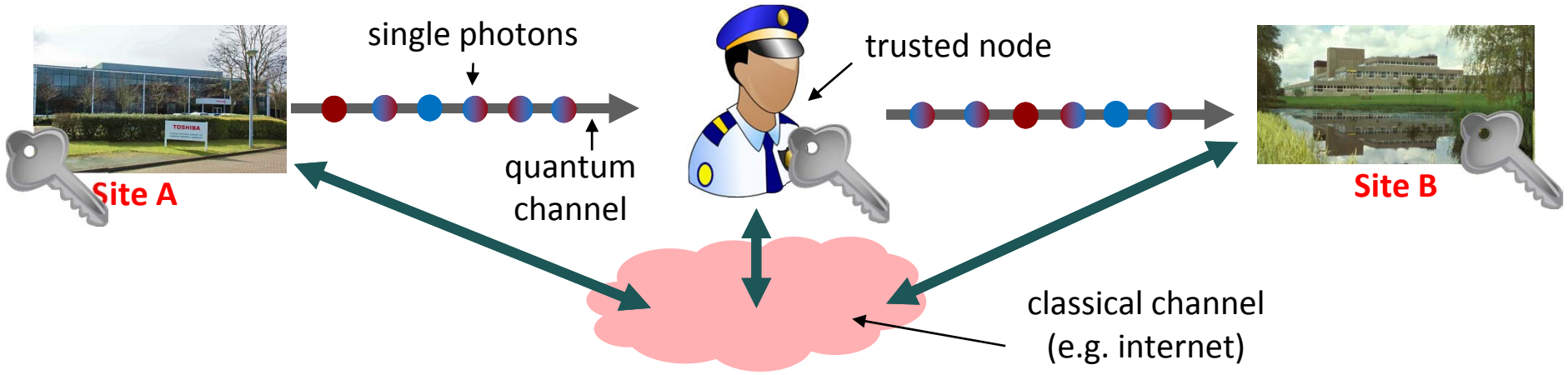
[3]Cavendish Laboratory, University of Cambridge, UK

Overview

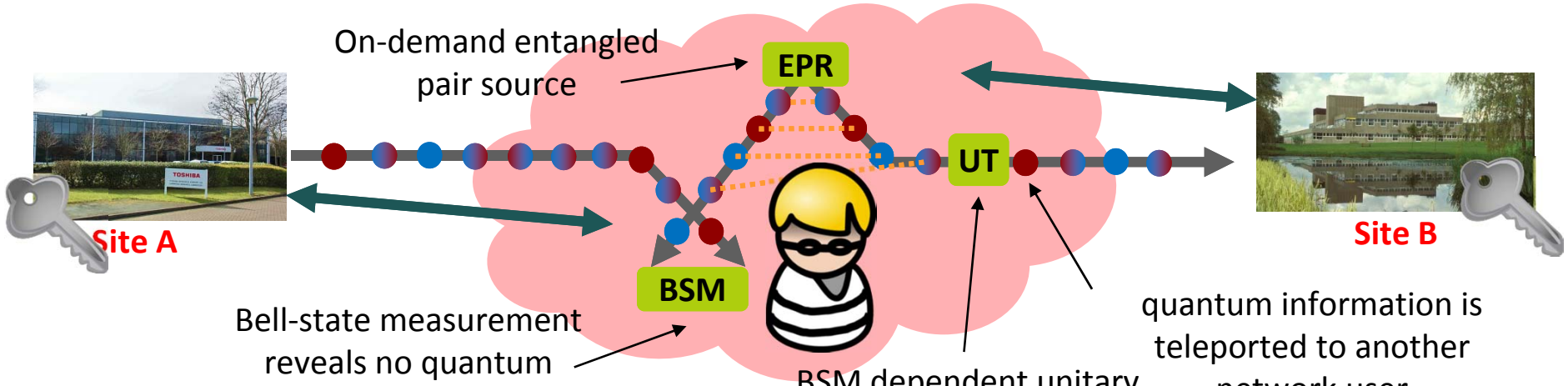
- **Introduction**
- **Entangled-LED (ELED) Design**
- **A quantum relay over 1km of optical fibre**
- **Results**

Secure quantum networks

- **Quantum networks with classical receive and send nodes require additional trust**



- **Quantum relays[1] are untrusted and can guarantee quantum network security**

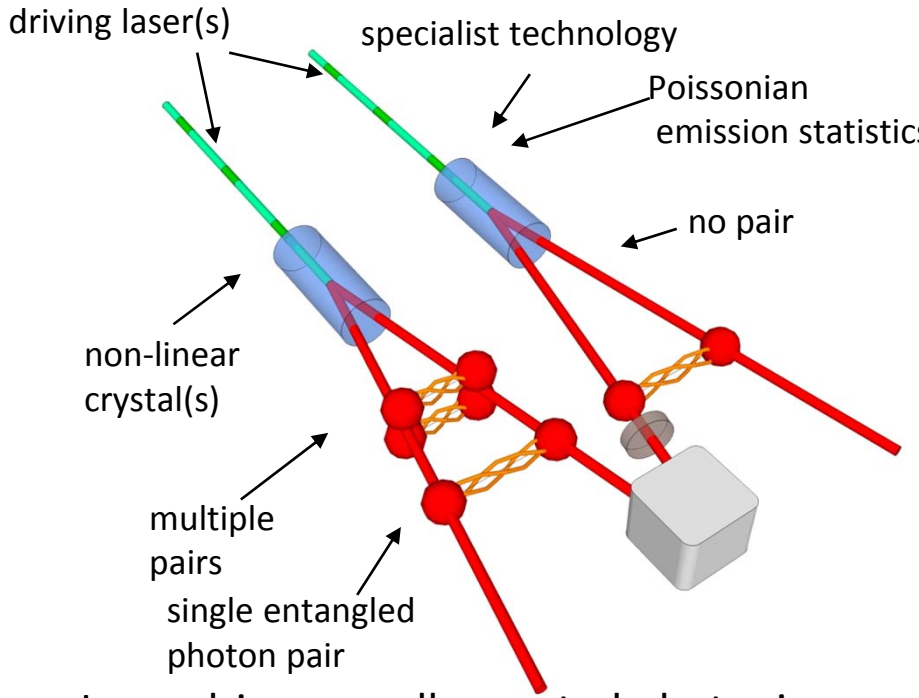


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[1] Jacobs, *Phys. Rev. A* **66**, 052307 (2002)

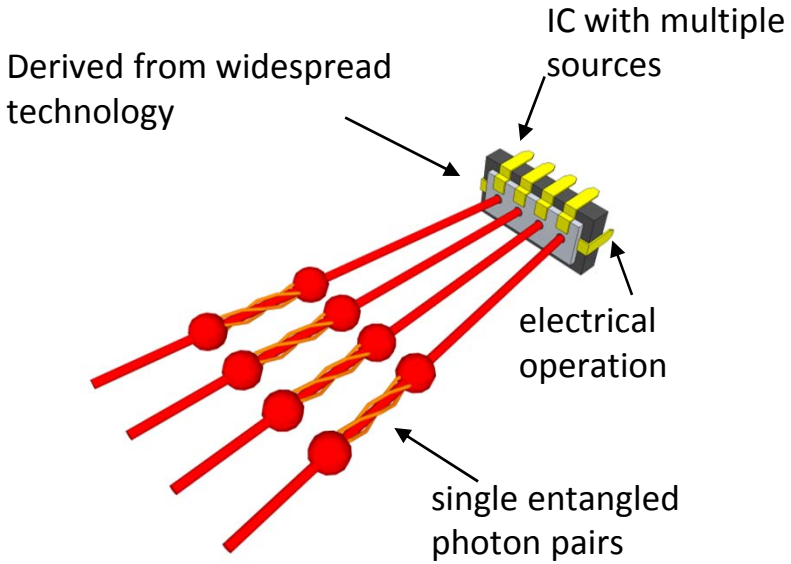
Photonic quantum teleportation platforms

Parametric Down Conversion [1,2]



- Laser driven, as all reported photonic teleportation
- Accidental coincidences (errors) can be very large, especially without heralding (e.g. 68% [1])

Quantum Dot Entangled-LEDs [3-5]



- Electrically driven semiconductor technology offers integration and practical scalability
- Sub-Poissonian emission prevents multi-pair emission, limits errors for large systems
- Less developed entangled technology

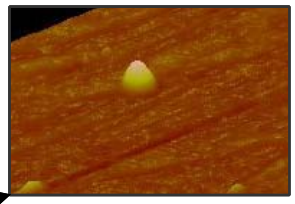
Scalable implementations benefit from sub-Poissonian, electrically-operated light sources

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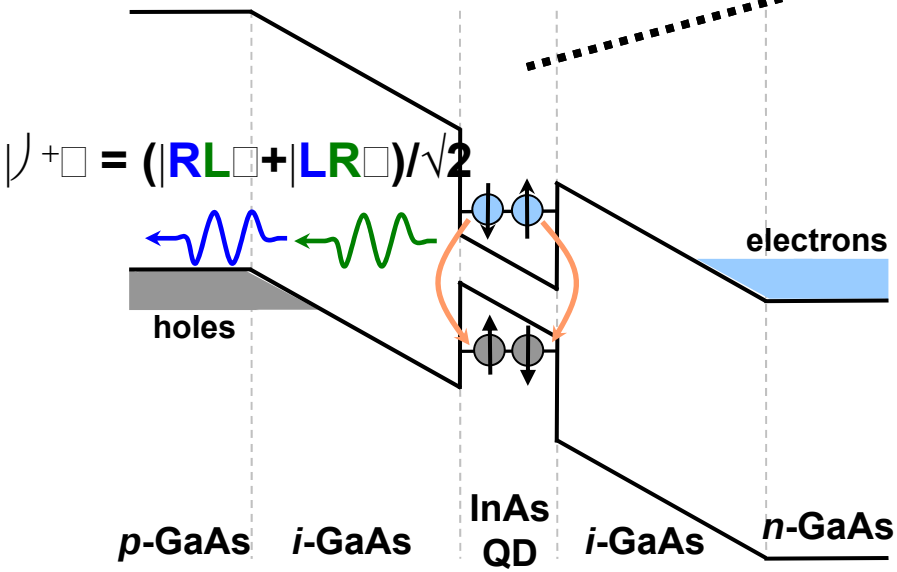
[1] Bouwmeester et al., Nature 390, 575 (1997), Ma et al., Nature 489, 269 (2012), [3] Benson et al., PRL 84, 2513 (2000), [4] Stevenson et al., Nature 439, 179 (2006), [5] Salter et al., Nature 465, 594 (2010)

Entangled photon pairs from the ELED

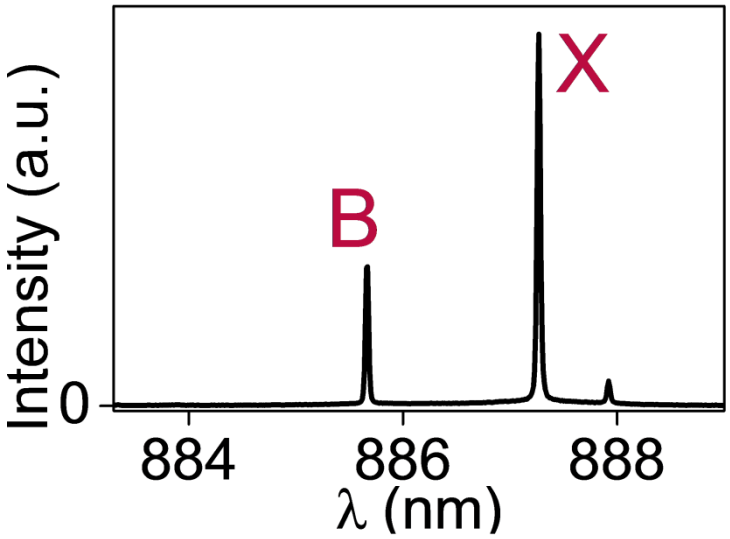
- LED p-i-n structure
- Biexciton cascade from a **quantum dot**
- 1 exciton (X) & 1 biexciton (B) photon
- single-photon emission



quantum dot
(15 x 5 nm)



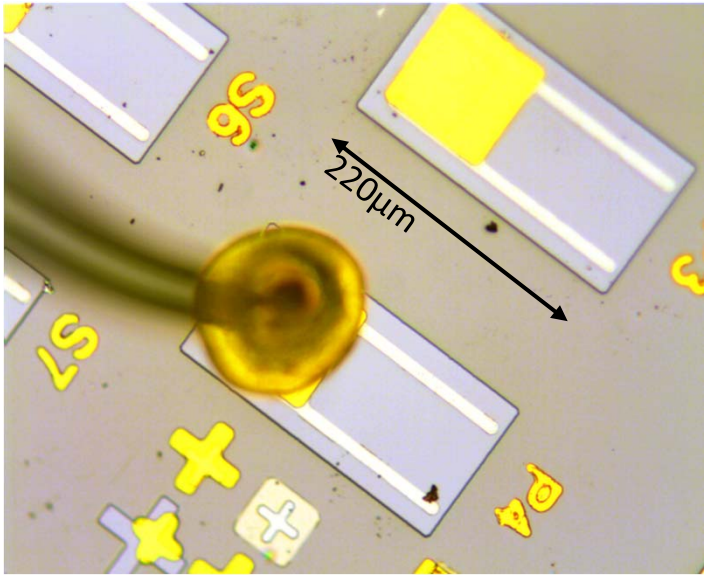
C.L. Salter et al, *Nature* **465**, 594 (2010).

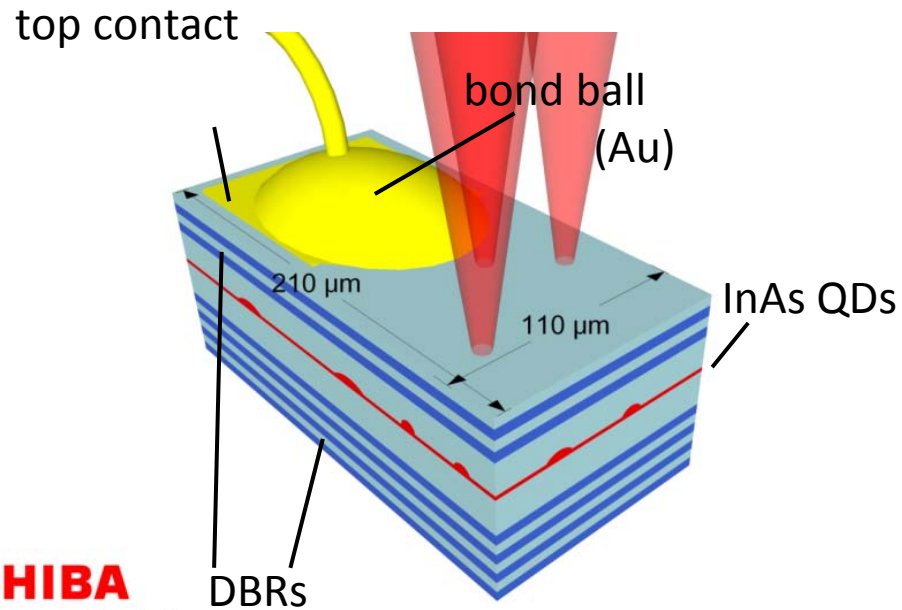


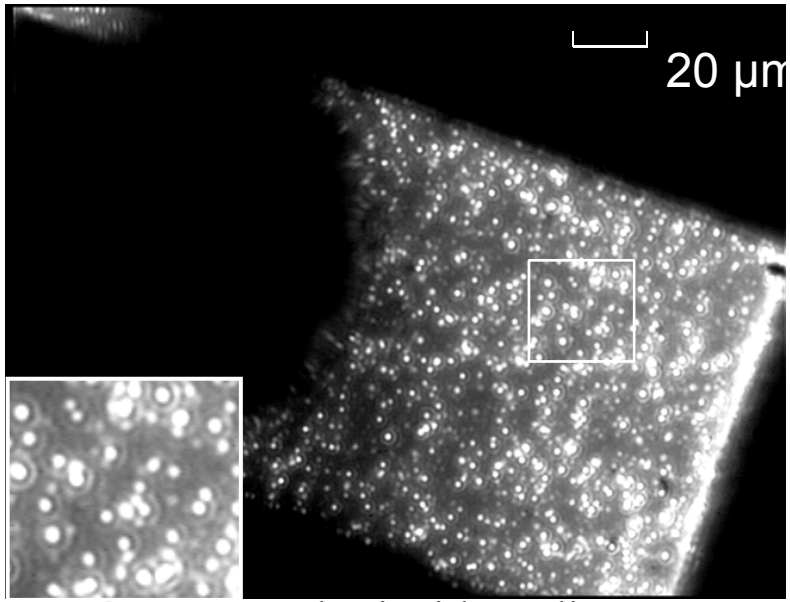
$$\Psi^+ = \frac{|RL\rangle + |LR\rangle}{\sqrt{2}} \equiv \frac{|HH\rangle + |VV\rangle}{\sqrt{2}} \equiv \frac{|DD\rangle + |AA\rangle}{\sqrt{2}}$$

E-LED Design

- Multiple devices on chip
- Individually bonded
- No apertures
- Small capacitance
- Allows for pulsed operation
- Sub-Poissonian emission





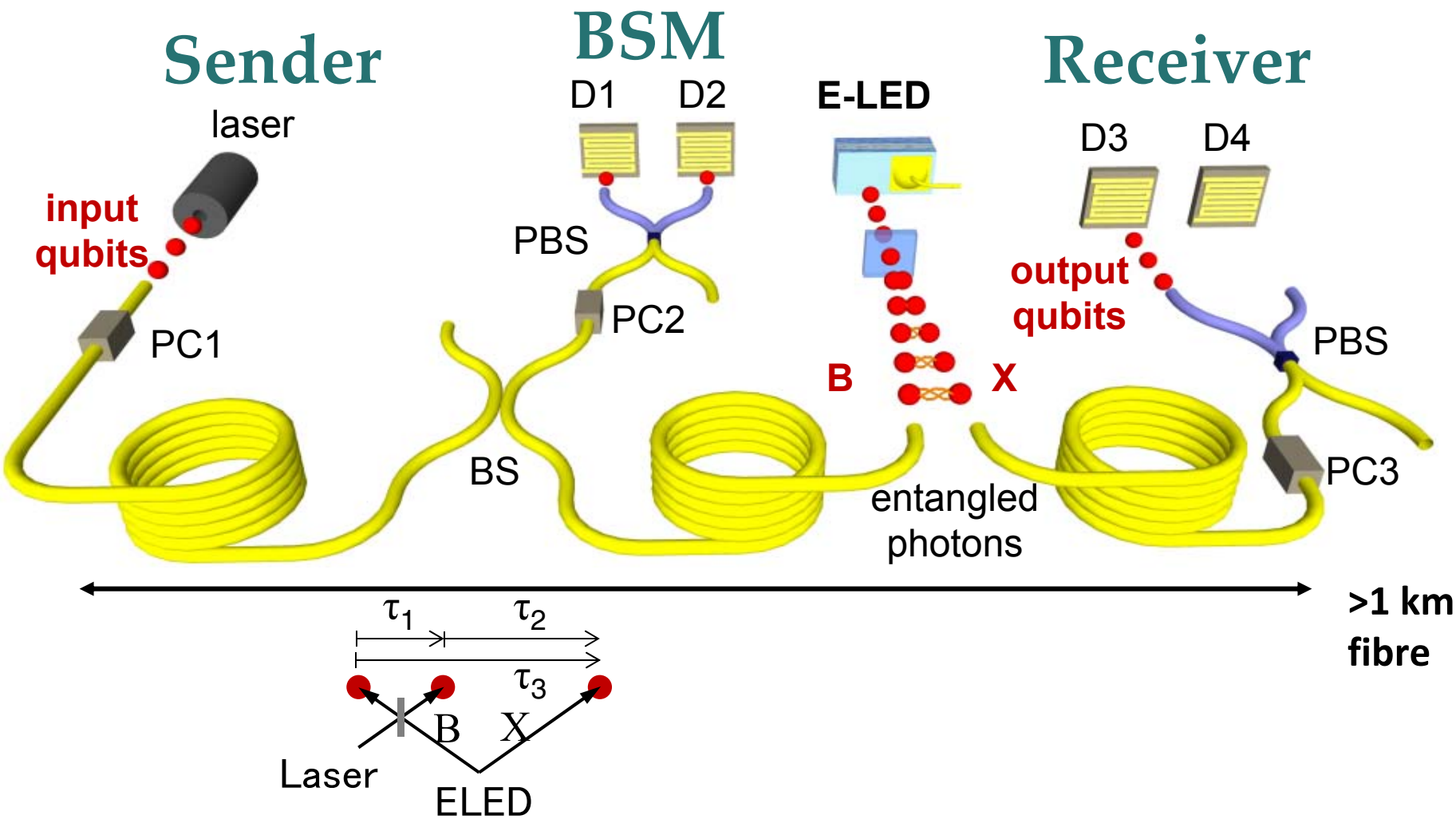


EL of quantum dot inside ordinary LED

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Quantum Relay over 1km of Optical Fibre

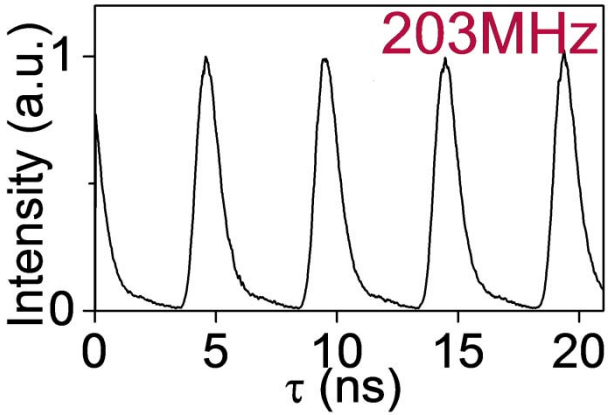
- Quantum Relay allows users on a network to communicate securely without trusting network nodes.



PC: Polarisation Controller, BS: Beamsplitter, PBS: Polarising Beamsplitter, SSPD: Superconducting Single Photon Detector, BSM: Bell State Measurement

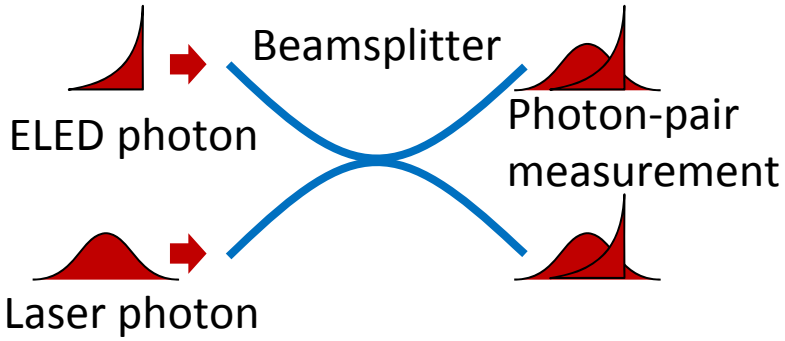
Experimental Details

- First pulsed teleportation experiment with ELED

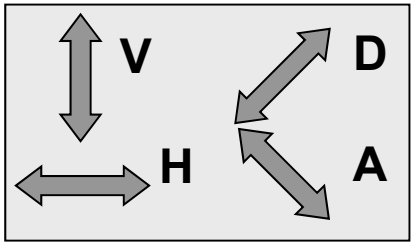


time-resolved EL of (B) emission

- qubit encoding: laser pulses tuned to dot wavelength & frequency

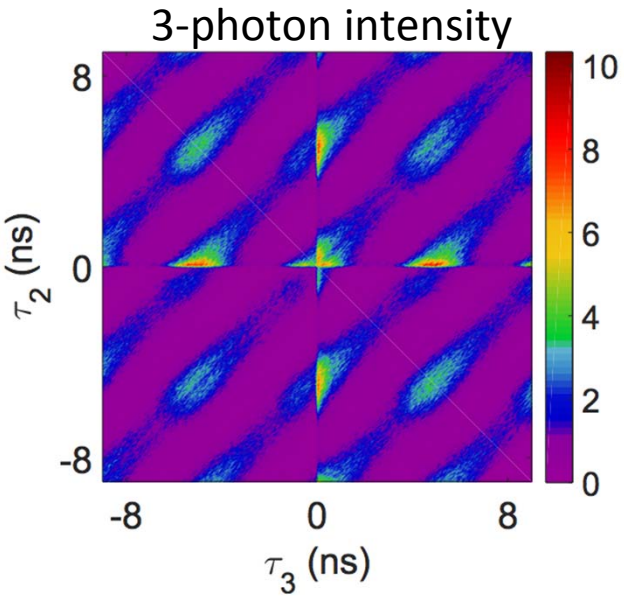


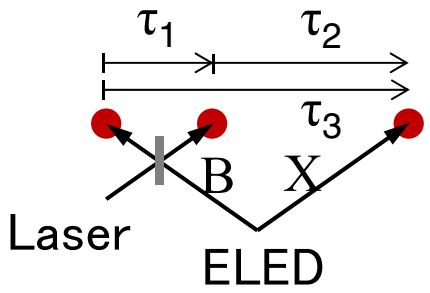
- polarization-encoded BB84 quantum states: D, A, H and V
- randomized sequence of teleported qubits was recorded

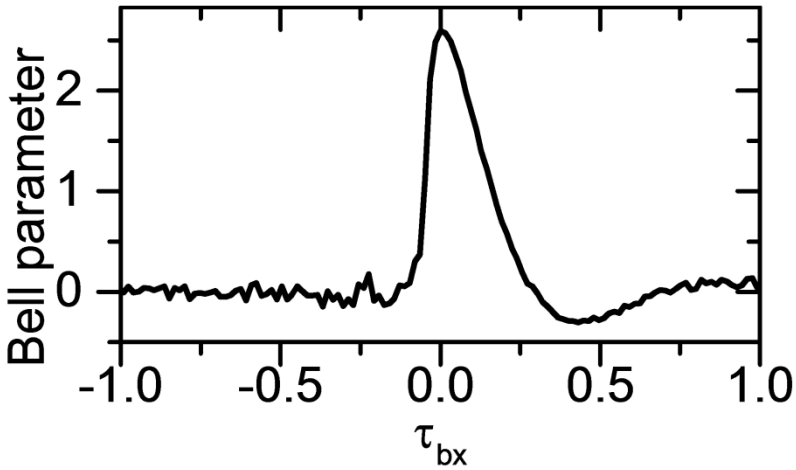


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Results





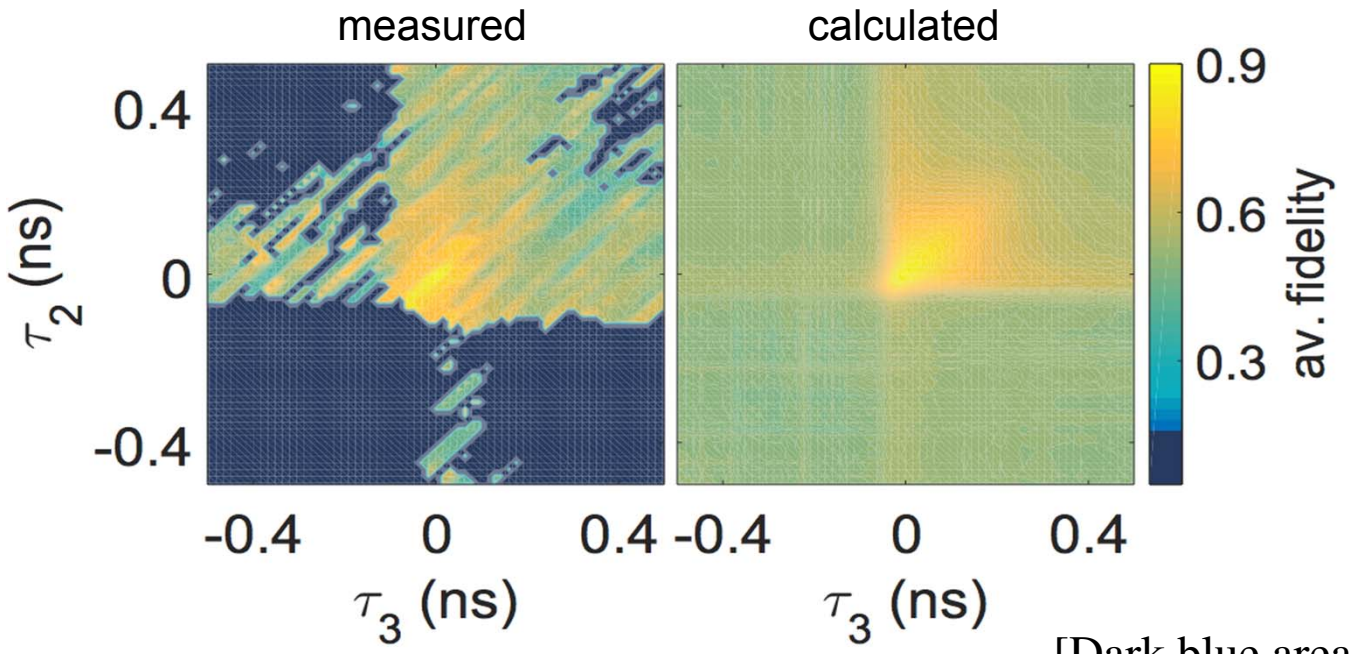


- Bell parameter 2.59±0.01 > 2
- Entanglement fidelity estimated ~0.94

- highly pulsed character
- Dip in coincidences at $\tau_2=\tau_3$:
  sub-Poissonian characteristic
- Advantage over the photon number splitting attack

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Results (2): Relay fidelity



[Dark blue area: no data]

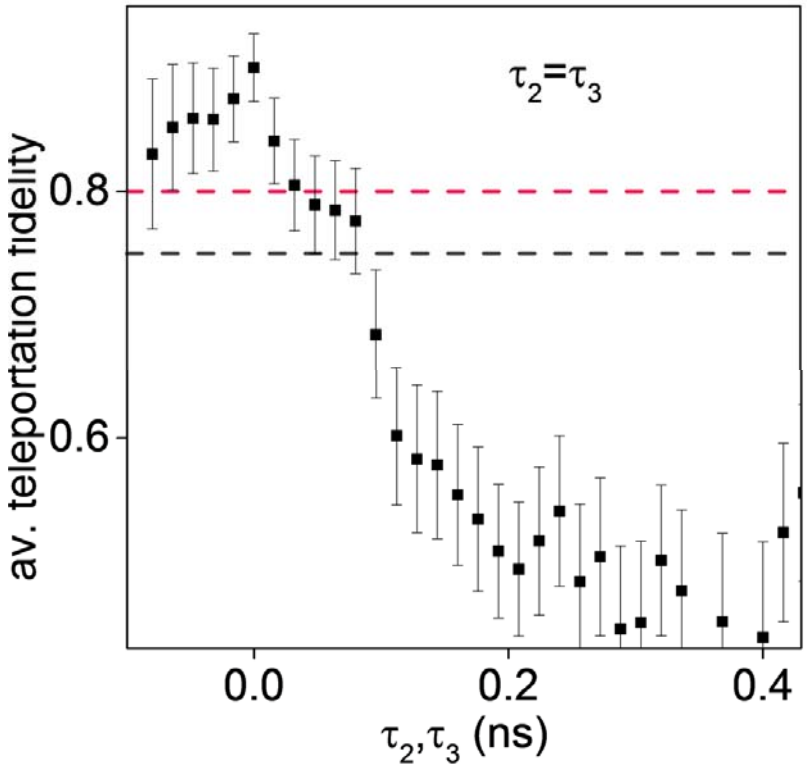
- Average relay fidelity f_{av} up to **0.900±0.028**

- Individual basis fidelities
  0.957± 0.042 (H)
  0.951±0.0475 (V)
  0.845± 0.064 (D)
  0.847±0.063 (A)

- two-photon interference between simultaneously detected photons is required for teleporting superposition states
- Limited by coherence time of photons

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Results (3): Relay fidelity



- Sender and Receiver have shared information in excess of any information held by an eavesdropper
- QBER = 1-f_{av} ~ 0.1 [f_{av}=0.90]
- f_{av} threshold for 4-state error correction[1] : 0.80 - *Well exceeded*
- Threshold for efficient 4-state BB84 protocol[2]: 0.89
- Secure key fraction: 0.11 bits per detected photon

[1] H. F. Chau, Phys. Rev. A 66, 060302(R) (2002)
[2]H.-K. Lo, H. F. Chau, and M. Ardehali, J. Cryptology **18**, 133 (2005)

Conclusions

- Entangled-LEDs are a practical quantum source with fundamental advantages
- First time entanglement has been distributed by a Q.D. source over a distance more than a few meters
- Average teleportation fidelity F exceeds classical limit and threshold for BB84 error correction protocols
- Quantum teleportation and quantum relays can be used to extend quantum networks beyond dedicated links
- ELED and teleportation performance sufficient for preliminary applications

Preprint: Varnava et al., arXiv:1506.00518 (2015)